「Section 5.1 If $f: \mathbb{R} \to \mathbb{R}$ satisfies that

$$|f(x) - f(y)| \le (x-y)^2 \quad \text{for all } x, y \in \mathbb{R}$$

Then f is constant.

Proof. Since $\left| \frac{f(x) - f(y)}{x-y} \right| \le |x-y|$ for all $x, y \in \mathbb{R}$,

given $x \in \mathbb{R}$, $f'(x)$ exists with $f'(x) = 0$.

Now, by the Mean-value Theorem, for any $a < b$, there exists $c \in (a,b)$ s.t

$$f(a) - f(b) = (a-b) \cdot f'(c) = 0$$

Thus f is constant.

「Section 5.2. Let $f: (a,b) \to \mathbb{R}$ be given. If $f'(x) > 0$ for all $x \in (a,b)$, then f is injective. Moreover, the inverse $f^{-1}: f[(a,b)] \to (a,b)$ exists and differentiable with

$$(f^{-1})'(f(x)) = \frac{1}{f'(x)} \quad (a < x < b)$$

Proof. Let $a < x < y < b$. Then, by the Mean-Value Theorem, there exists $c \in (x,y)$ s.t

$$f(y) - f(x) = f'(c) \cdot (y - x) > 0$$

Therefore, f is strictly increasing, thus injective. Now, the restriction

$$f: (a,b) \to f[(a,b)]$$

is bijection, thus there exists a inverse

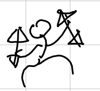
$$f^{-1}: f[(a,b)] \to (a,b) : f(x) \mapsto x$$

Now, $(f^{-1} \circ f)(x) = x$ implies that

⌈Section 5.3. Let $g : \mathbb{R} \to \mathbb{R}$ be given with $|g'| \le M$ for some $M > 0$.
Fix $\varepsilon > 0$, and define $f(x) = x + \varepsilon \cdot g(x)$. prove that f is injective if ε is small enough.
Proof. If $\varepsilon < \frac{1}{M}$, then $f'(x) = 1 + \varepsilon \cdot g'(x) > 0$. Therefore f is injective.

⌈Section 5.4. If $C_0, \cdots, C_n \in \mathbb{R}$ satisfies

$$C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \cdots + \frac{C_{n-1}}{n} + \frac{C_n}{n+1} = 0$$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, then the polynomial

$$C_0 + C_1 x + C_2 x^2 + \cdots + C_{n-1} x^{n-1} + C_n x^n = p(x)$$

has a real root in $(0,1)$.
Proof. Define $$g(x) = C_0 x + \frac{C_1}{2} x^2 + \cdots + \frac{C_{n-1}}{n} x^n + \frac{C_n}{n+1} x^{n+1}$$

Then, $g(0) = 0$ clearly, and $g(1) = 0$ by the assumption.
By the Mean-Value Theorem, there exists $\alpha \in (0,1)$ s.t

$$0 = g(1) - g(0) = g'(\alpha) \cdot (1-0) = g'(\alpha)$$

Directly, $p(x) = g'(x)$. therefore $g'(\alpha) = p(\alpha) = 0$.

⌈Section 5.5 Let $f : (0,\infty) \to \mathbb{R}$ be differentiable.
If $f'(x) \to 0$ as $x \to \infty$, then $f(x+1) - f(x) \to 0$ $x \to \infty$.
Proof. By the MVT, for any $x > 0$, there exists $C_x \in (x, x+1)$ s.t

$$f(x+1) - f(x) = f'(C_x) \cdot (1-0) = f'(C_x).$$

Let $\varepsilon > 0$ be given. Then, there exists $\delta > 0$ s.t $x \ge \delta \Rightarrow |f'(x)| < \varepsilon$.
For this $\delta > 0$, for any $x \ge \delta$, there is $C_x \in (x, x+1)$ s.t
$$|f(x+1) - f(x)| = |f'(C_x)| < \varepsilon$$
Thus proved.

Section 5.6. Suppose that $f:(0,\infty)\to\mathbb{R}$ satisfies that;

1). f is Continuous for $x\geq 0$
2). f' exists for $x>0$
3). f' increases monotonically.
4). $f(0)=0$.

Then, $g:(0,\infty)\to\mathbb{R}: x\mapsto \frac{f(x)}{x}$ increases monotonically.

Proof. Since $g'(x)=\dfrac{xf'(x)-f(x)}{x^2}$, enough to show $f'(x)\geq \frac{f(x)}{x}=g(x)$.

observe that $g(x)=\dfrac{f(x)}{x}=\dfrac{f(x)-f(0)}{x-0}=f'(c_x)$ for some $0<c_x<x$, by the MVT.

Now, Since f' increases monotonically $f'(x)\geq g(x)=f'(c_x)$.

Section 5.7. Let $f,g:\mathbb{R}\to\mathbb{R}$ be given.

Suppose that f', g' exist. and $g'(x)\neq 0$ for all $x\in\mathbb{R}$, and $f(x)=g(x)=0$.

Prove $\lim\limits_{t\to x}\dfrac{f(t)}{g(t)}=\dfrac{f'(t)}{g'(t)}$

Proof. Since f', g' exist, and $f(x)=g(x)=0$,

$$\lim_{t\to x}\frac{f(t)}{g(t)}=\lim_{t\to x}\frac{\frac{f(t)-f(x)}{t-x}}{\frac{g(t)-g(x)}{t-x}}=\frac{f'(x)}{g'(x)}$$

$\sum\limits_{n=1}^{\infty}\sin\left(\frac{1}{n}\right)$ Converges?

$\dfrac{\sin\frac{1}{n}}{\frac{1}{n}}\to 1$ but $\sum\frac{1}{n}$ diverges, thus

⌈Section 5.8. Let $f : [a,b] \to \mathbb{R}$ be given. Suppose f' exists and continuous on $[a,b]$.
Given $\varepsilon > 0$, there exists $\delta > 0$ s.t. $\forall x, t \in [a,b]$, $|x-t| < \delta \Rightarrow \left| \dfrac{f(x) - f(t)}{x - t} - f'(x) \right| < \varepsilon$

「Section 5.9.

$$f: \mathbb{R} \to \mathbb{R}; x \mapsto \begin{cases} x+3 & x \geq 0. \\ 3 & x < 0 \end{cases}$$

「Section 5.10. Let $f, g: (0,1) \to \mathbb{C}$ be differentiable such that $\lim_{x\to 0} f(x) = \lim_{x\to 0} g(x) = 0$ and $\lim_{x\to 0} f'(x)$, $\lim_{x\to 0} g'(x) (\neq 0)$ exist. Then,

$$\lim_{x\to 0} \frac{f(x)}{g(x)} = \lim_{x\to 0} \frac{f'(x)}{g'(x)}$$

Proof. observe that $\dfrac{f(x)}{g(x)} = \left[\dfrac{f(x)}{x} - \lim_{t\to 0} f'(t) \right] \cdot \dfrac{x}{g(x)} + \lim_{t\to 0} f'(t) \cdot \dfrac{x}{g(x)}$

Meanwhile, by the L'Hopital Law,

$$\lim_{x\to 0} \frac{f(x)}{x} = \lim_{x\to 0} \left[\frac{\operatorname{Re} f(x)}{x} + i \cdot \frac{\operatorname{Im} f(x)}{x} \right]$$

$$= \lim_{x\to 0} \left[\operatorname{Re} f'(x) + i \cdot \operatorname{Im} f'(x) \right] = \lim_{x\to 0} f'(x)$$

Similarly $\lim_{x\to 0} \dfrac{x}{g(x)} = \left(\lim_{x\to 0} g'(x) \right)^{-1}$. Now,

$$\lim_{x\to 0} \frac{f(x)}{g(x)} = \left(\lim_{x\to 0} f'(x) \right) \cdot \left(\lim_{x\to 0} g'(x) \right)^{-1}$$

「Section 5.11.

Suppose that f is defined on a neiborhood of $x \in \mathbb{R}$, and $f''(x)$ exists. Prove that

$$\lim_{h \to 0} \frac{f(x+h)+f(x-h)-2f(x)}{h^2} = f''(x)$$

Proof. By the L'Hopital Law,

$$\lim_{h\to 0} \frac{f(x+h)+f(x-h)-2f(x)}{h^2} \overset{①}{=} \lim_{h\to 0} \frac{f'(x+h)-f'(x-h)}{2h} \overset{②}{=} \frac{1}{2}\cdot \lim_{h\to 0}\left(f''(x+h)+f''(x-h)\right) = f''(x)$$

But, if $f''(x)$ does not exist, then we can not apply L'Hopital Law in ②.

Define $f : \mathbb{R} \to \mathbb{R} : x \mapsto \begin{cases} 0 & \text{if } x<0 \\ x^2 & \text{if } x \ge 0 \end{cases}$

Then, $f''(0)$ does not exists, But

$$\lim_{h \to 0} \frac{h^2+0-0}{h^2} = 1$$

exists.

「Fail Try

By the Mean-Value Theorem, for any $h>0$,

$f(x+h)-f(x) = f'(t_1)h$ and $f(x-h)-f(x) = f'(t_2)h$ for some

for some $t_1 \in (x, x+h)$, $t_2 \in (x-h, x)$

$$\frac{f'(t_1)+f'(t_2)}{h} = 2f''(x)$$